The Research Lifecycle in Three Supply Chain Resilience Dissertations

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The research lifecycle runs from identifying a problem through design, data collection, analysis, interpretation, and dissemination, with ethical obligations carried through every stage. This paper reviews three dissertations that address supply chain disruption and resilience, a topic directly related to my proposed study, *Mitigation of Supply Chain Uncertainties Caused by Material Shortage During Ramp-Up*, which focuses on semiconductor manufacturing equipment providers and their suppliers: a qualitative multiple case study (Benton, 2019), a quantitative survey study (Ponomarov, 2012), and a multiphase mixed methods study (Pettit, 2008). A closing synthesis draws lessons for the proposed study.

Qualitative Dissertation: Benton (2019)

Research Problem, Purpose, and Questions

Benton (2019) addressed the problem that supply chain disruptions erode manufacturers' profitability and that some managers lack effective mitigation strategies. The purpose was to explore the strategies manufacturing managers use to mitigate disruptions to remain profitable. The central research question asked what strategies manufacturing managers use to mitigate supply chain disruptions to remain profitable. Consistent with exploratory qualitative inquiry, the study stated no hypotheses.

Methodology and Design

Benton (2019) selected a qualitative methodology because the goal was to understand how practitioners experience and respond to disruptions, not to test relationships among variables. The design was a multiple case study of two manufacturing firms in eastern Virginia, with four purposively selected supply chain managers who had successfully implemented

mitigation strategies. Contingency theory of fit, which links performance to aligning practices with environmental conditions, framed the study.

Research Instruments and Analysis

The researcher was the primary instrument. Data came from semistructured interviews guided by an interview protocol, company documents, and publicly available information about the participating firms. Using several data sources allowed methodological triangulation. Benton (2019) used thematic analysis: transcripts and documents were coded, codes were grouped and compared across participants and firms until recurring themes emerged, and the themes were interpreted through contingency theory and the supply chain risk literature.

Key Findings

Three themes emerged: (a) collaboration and information sharing, (b) information technology and supply chain risk, and (c) the use of multiple suppliers (Benton, 2019). Managers reduced disruption risk by sharing information with partners, using technology to increase visibility, and avoiding single-source dependence.

Ethical and Legal Implications

Research with human participants requires institutional review board (IRB) approval, informed consent, and the right to withdraw. Small samples from identifiable firms risk reidentification, so pseudonyms, secure storage, and confidentiality for people and organizations are essential. Legally, company documents may contain proprietary terms covered by nondisclosure agreements, which limits how they can be quoted.

Quantitative Dissertation: Ponomarov (2012)

Research Problem, Purpose, Questions, and Hypotheses

Ponomarov (2012) addressed a gap: resilience had been discussed conceptually, but little empirical evidence showed which firm capabilities produce it or whether it creates measurable value. The purpose was to examine the antecedents and value-based consequences of supply chain resilience from the perspective of the individual firm. The guiding questions asked which organizational orientations and capabilities lead to resilience and what outcomes resilience produces for the firm. The hypothesized model, grounded in a dynamic capabilities extension of the resource-based view, proposed directional relationships linking supply chain orientation and risk management orientation to logistics capabilities, logistics capabilities to supply chain resilience, and resilience to firm-level consequences.

Methodology and Design

The methodology was quantitative and deductive, deriving hypotheses from theory and testing them statistically. The design was a cross-sectional, nonexperimental survey of supply chain and logistics managers (Ponomarov, 2012), estimating relationships among constructs across many firms.

Research Instruments and Analysis

The primary instrument was a structured questionnaire made up of multi-item measurement scales for each construct, including supply chain orientation, risk management orientation, logistics capabilities, and supply chain resilience (Ponomarov, 2012), allowing reliability and validity to be assessed statistically. Ponomarov (2012) analyzed the data with structural equation modeling (SEM). SEM proceeds in two stages: assessing the measurement model to confirm that each scale is reliable and that constructs show convergent and discriminant

validity, then estimating the structural model to test the hypothesized paths, including indirect effects, simultaneously.

Key Findings

The results supported a capability-based explanation of resilience. Firms' orientations toward supply chain management and risk management influenced the logistics capabilities they developed, and those capabilities contributed to supply chain resilience, which in turn carried value-based consequences for the firm (Ponomarov, 2012).

Ethical and Legal Implications

Survey research with professionals requires IRB review, informed consent, and protection of respondent anonymity, particularly when questions touch on competitively sensitive practices. Researchers must also test for common method bias and report unsupported hypotheses transparently. Legally, electronically collected responses must comply with data privacy requirements.

Mixed Methods Dissertation: Pettit (2008)

Research Problem, Purpose, Questions, and Propositions

Pettit (2008) addressed the absence of an integrated, measurable framework for assessing and improving supply chain resilience. The purpose was threefold: to develop a conceptual framework of resilience, to create and validate an assessment tool, and to identify an implementation process. The research questions asked what constitutes supply chain resilience, whether it can be measured reliably, and which linkages between resilience factors should guide improvement. In place of formal statistical hypotheses, the framework advanced propositions, notably that firms should seek a balanced zone in which capabilities match vulnerabilities without excessive cost.

Methodology and Design

The methodology was mixed methods, with a multiphase sequential design (Pettit, 2008). Phase I built the conceptual framework from the literature and refined it with practitioner focus groups (qualitative). Phase II developed and validated a resilience assessment tool by comparing quantitative assessment scores with qualitative data from focus groups at participating global manufacturing and service firms that reviewed recent disruptions. Phase III identified critical linkages among resilience factors.

Research Instruments and Analysis

The instruments included focus group protocols for eliciting practitioner experience with disruptions and a structured self-assessment questionnaire built from the attributes of the framework. Respondents rated their firm's exposure on vulnerability attributes and its strength on capability attributes (Pettit, 2008). Qualitative focus group data were coded and mapped to framework factors, and quantitative assessment results were scored and examined through correlations. Pettit (2008) then used triangulation across theoretical linkages, assessment correlations, and focus group connections, which is the defining integration step of a mixed methods study, to determine which capabilities addressed which vulnerabilities.

Key Findings

Pettit (2008) found that resilience can be assessed along two dimensions, vulnerabilities and capabilities. The study identified seven vulnerability factors comprising 40 attributes and 14 capability factors comprising 71 attributes. Triangulation produced 311 specific linkages that leaders can use to prioritize resilience investments, and the assessment tool was validated.

Ethical and Legal Implications

Research inside corporations requires keeping firm identities and disruption histories confidential and reporting in aggregate. Focus groups limit confidentiality because participants hear one another, and university-industry collaboration requires managing conflicts of interest. Legally, the assessment tool later developed into a trademarked instrument, which raises intellectual property questions about ownership and the licensing of research products.

Comparative Synthesis and Implications for the Proposed Study

The research lifecycle changes with methodology. Benton (2019) began with a practical question and built understanding inductively from a small number of information-rich participants. Ponomarov (2012) began with theory, derived hypotheses, and tested them deductively across a broader sample. Pettit (2008) combined both logics. The findings converge. Supplier collaboration, information visibility, multiple sourcing, flexibility, and deliberate capability building appear across all three, and these are the same mitigation levers named in my proposed framework.

For the proposed qualitative case study of semiconductor ramp-up, Benton's (2019) design offers a direct model for semistructured interviews, document review, and thematic analysis with equipment providers and their suppliers. Ponomarov's (2012) constructs can inform interview coding categories, and Pettit's (2008) vulnerability-capability lens can structure analysis of material shortages, long lead times, and shifts in supplier-buyer power. The semiconductor context sharpens ethical and legal concerns: participants' firms operate under strict nondisclosure agreements, and equipment and materials data may be subject to export control regulations, so IRB approval, informed consent, de-identification of firms, and careful screening of proprietary documents will be central to the proposed research.

References

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